



Final Project

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1 Overview

The goal of this project is for you to further explore a certain subject that we covered in class or a new subject related to it (based on some previous work). This is an open project in a sense that you will propose the problem you would like to work on and the solution you are planning to use. Your main goal should be to explore the characteristics of your problem/solution while using LLMs.

2 A Few Guidelines

1. Most projects fall into one of these three categories:
(taken from <http://cs229.stanford.edu/projects.html>)
 - **Application Project:** This is by far the most common: Pick an application that interests you and explore how best to apply LLMs/agents to solve it.
 - **Algorithmic project:** Pick an algorithm or a family of algorithms and explore an interesting aspect of it through extensive experimentation.
 - **Theoretical project:** Prove some interesting/non-trivial properties of a new or an existing AI algorithm. (This is often quite difficult, and so very few, if any, projects will be purely theoretical.)
2. Your work should be novel in some respect. That is, it should either address a new domain or explore certain algorithms in a specific domain.
3. Your project will not be judged by how well it succeeded, but rather by how well it was motivated, thought of and tested. Therefore, feel free to try some far fetched ideas.
4. In general, you should not propose a completely new idea. Rather, you should further develop an issue explored in the course or from a paper, or the combination of two issues.

5. Try to find a recent paper which addresses the topic of your choice, and relate your work to it.
6. Try to not introduce multiple novelties - try to focus on one aspect so you can explore it sufficiently.
7. Make sure that the question you bring up in your paper, is answered by your paper.
8. Feel free to integrate things you are already working on into your project.
9. Good previous work can be found at the ICML and NeurIPS conferences.
10. Depending on your work you will be using local or cloud based LLMs
11. Teams can be 3-4 students.

3 Sample Projects

- Applications:
 - *Build your own agent*
Build an autonomous agent to solve a problem in a specific domain. The agent must include the use of multiple tools and you must come up with some sort of evaluation policy. It would be good to compare multiple versions of the agent with different capabilities.
 - *LLMs for games*
Choose a game and evaluate how well an LLM can do at this game. Try different improvements using fine tuning, agents, etc.
- Algorithmic:
 - *RAG vs. Agentic AI* Compare traditional RAG with an agentic approach on a specific benchmark. This comparison should be not only focused on performance but time/computation/price as well.
 - *Small vs. Large Language Models*
Compare the performance of small and large language models on a specific task. Show that for specific tasks it may be enough to use a small model. Conduct extensive experimentation.
 - *The effects of long context windows*
Compare different attention mechanisms on their ability to retrieve specific facts buried in the middle of a long token context. How does the location of the text effect the results? What about different models?
 - *The use of reasoning*
Compare different models on how reliant they are on reasoning outputs. What happens when you alter those? How does this depend on size?

4 Coding(30%)

You are required to write your code in python3. You are allowed to use any external libraries.

In addition, you may use other publicly available code or AI generated code - given that you cite it, and describe the exact changes you made to it.

5 Report (70%)

I will upload a Latex template to Canvas. Your report will need to include the following sections:

1. Abstract(5%)

Make sure to clearly state the novelty of your work in the abstract.

2. Introduction(5%)

In this section you should discuss the question you are planning to investigate. Make sure to describe the problem clearly, and how you are planning to solve it. Also, clarify what your new contribution is (as opposed to previous works).

3. Previous Work(10%)

Discuss the previous work you are using as a starting point/reference. If you are working on a different domain include work that works on the same or similar domain. If you are using a new method, discuss other works which use similar methods. Note that you can use both academic papers in addition to online articles. Make sure to cite all sources.

4. Technical Approach (10%)

Describe the problem you are working on including the rules, goals, etc. Assume that the reader has not been introduced to this domain problem previously. Describe your solution architecture. If you are examining different options describe them all. In addition, add diagrams to clarify.

5. Dataset and Implementation (10%)

Describe the details of your dataset (if applicable) and implementation. This should include things like the size of your dataset, libraries used, hardware used, etc.

6. Experiments and Results Analysis(20%)

Present the results which answer your hypothesis questions. These should include different graphs which clarify the answer and examine different aspects of your solution. If not described in the implementation section, make sure to carefully describe how your experiments were conducted (how many runs, how it was initialized, etc. This section is the most important, and will be evaluated both on presentation and correctness.

7. Conclusion(5%)

What have you achieved in this project? What have you learned?

8. References

9. Appendices:

- Code Design(5%)

Describe the general structure of your code including functions, classes, and data structures used. If you are using code written by someone else make sure to cite it and emphasize the parts that you have written yourself.

- Workload Distribution

Describe how work was divided between the team members

6 Submission

You are required to submit two files. One pdf file of the report and the other a zip file with your code.